

財政學 HW3

1. (Hyman 11.3) The price elasticity of demand for wine is estimated to be -1 at all possible quantities. Currently, 200 million gallons of wine are sold per year, and the price averages \$6 per bottle. Assuming that the price elasticity of supply of wine is 1 and the current tax rate is \$1 per bottle, calculate the current excess burden of the tax on wine. Suppose the tax per bottle is increased to \$2 per bottle. What will happen to the excess burden of the tax as a result of the tax increase? Under what circumstances can a doubling of the tax on wine actually improve resource use in the United States, despite the increase in the excess burden of the tax?

2. (Hyman 11.5) Suppose the supply of housing construction is infinitely elastic at a price of \$150 per square foot. Currently 1 million square feet are built per month. If the price elasticity of demand for housing is -1, calculate the monthly excess burden of a 10 percent tax on housing construction. (Hint: Go to the discussion on taxation of constant cost industries.) What is the monthly excess burden if the tax is 20 percent? Who will bear the incidence of the tax?

3. (Rosen 15.1) Which of the following is likely to impose a large excess burden?

- a. A tax on land.
- b. A tax of 24 percent on the use of cellular phones. (This is the approximate sum of federal and state tax rates in California, New York, and Florida.)
- c. A subsidy for investment in “high-tech” companies.
- d. A tax on soda bought in a cup or glass but not bought in a bottle or can. (Such a tax exists in Chicago.)
- e. A 10-cent tax on a deck of cards. (Such a tax exists in Alabama.)
- f. A tax on blueberries. (Such a tax exists in Maine.)

4. (Rosen 15.9) In the United Kingdom, each household that owns a television pays a compulsory levy that is equivalent to \$233 per year. The total revenue collected, which is over \$7 billion annually, goes to the British Broadcasting Corporation. Do you think that such a tax is likely to have a substantial excess burden relative to the revenues collected?

5. (Rosen 15.12) In an effort to reduce alcohol consumption, the government is considering a \$1 tax on each gallon of liquor sold (the tax is levied on producers). Suppose that the supply curve for liquor is upward sloping and its equation is $Q=30,000P$ (where Q is the number of gallons of liquor and P is the price per gallon).

The demand curve for liquor is $Q=500,000-20,000P$.

- a. Draw a sketch to illustrate the excess burden of the tax. Next use algebra to calculate the excess burden. Show graphically the excess burden generated by the \$1 unit tax. (Hint: Compare the losses of both consumer and producer surplus to tax revenues.)
- b. Suppose that each gallon of liquor consumed generates a negative external cost of \$0.50. How does this affect the excess burden associated with the unit tax on liquor?